Pintos Overview

—— Why, What and How

TA Session

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PPT Credit: 向昱行 仲殷旻

Deadlines

Two deadlines: code dealines & design doc deadlines.
Check https://pku-os.github.io//sp26/ frequently.

1 week for lab0
2 week for lab1,2,3a,3b

Welcome to the World of Operating System

Educational OS Project Zoo

JOS

IA32
MIT6.828

xv6

RISCV32
MIT6.S081

Nachos

MIPS
Old CS162

Pintos

IA32
CS162

Q : Why Pintos ?

Design and Implementation

- OSDI, NSDI, PLDI … …
- Your design matters !!

- Talk is cheap, show me the code
- Write 2000+ LOC in a 10000+ LOC codebase

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Q : Why Pintos ?

You will learn by Read The Code

- important skill both in production and research
- learn from good coding style
- some tools may help you

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Q : Why Pintos ?

You will learn by Design The Code

- think tenth, code once
- design doc template may help you
- not Pintos, but Your Pintos

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Q : Why Pintos ?

You will learn by Write The Code

- maybe your first time writing 2000+ LOC
- tricky multi-threading synchronization
- test-driven development

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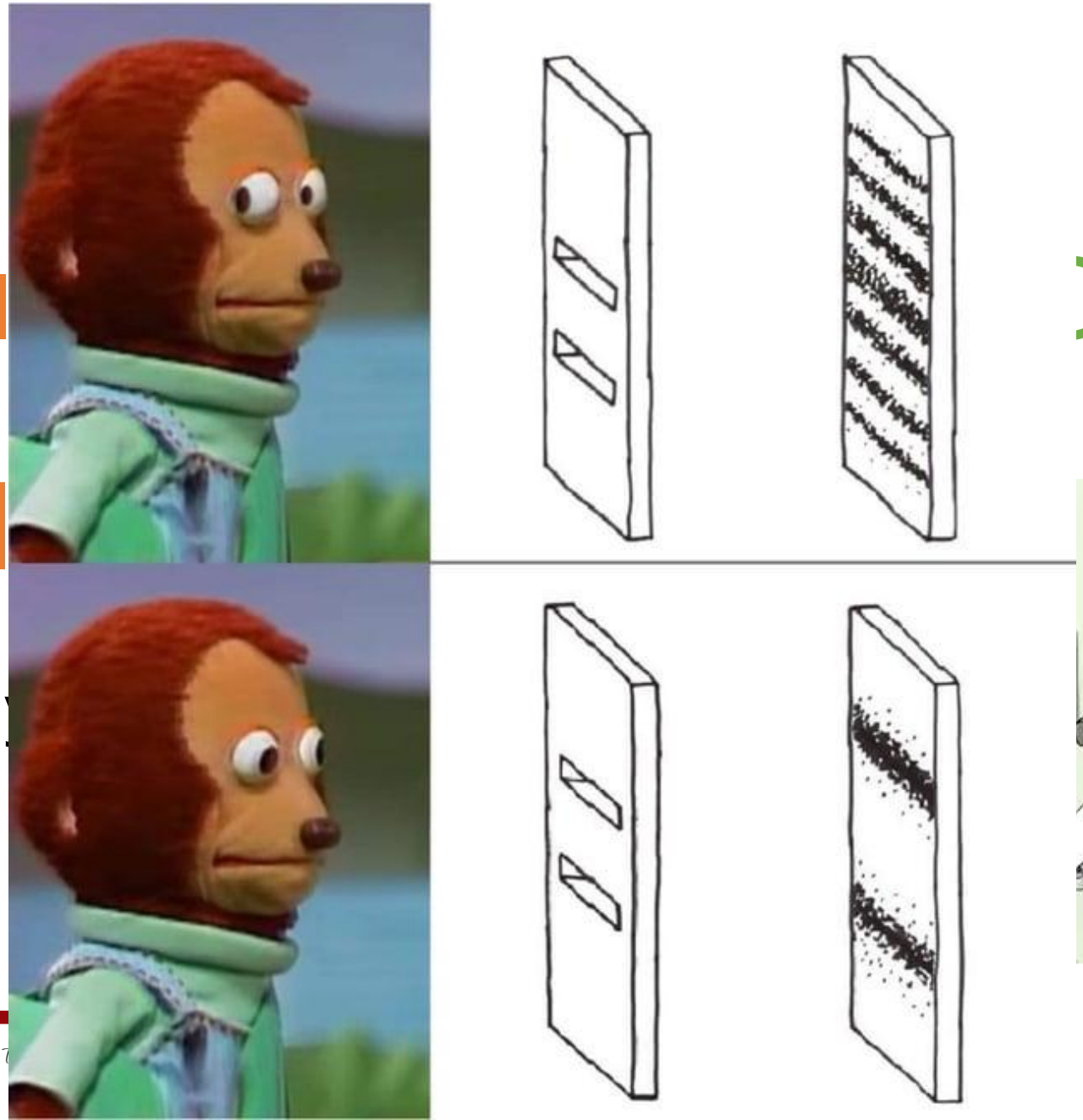
Me, debugging

Q : Why

You wil

Code

- You will
- start earl



Q : Why not Pintos ?

- IA32 architecture : CISC ISA, historical legacy

Pintos → Tacos

TACOS

RUST OS NEVER GETS RUSTY.

Pintos reimplemented in Rust based on RISCV64.

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Q : Why not Pintos ?

- IA32 architecture : CISC ISA, historical legacy

Pintos → Tacos

- time consuming : 100 hours +++

optional lab4, long long long lab document, per-lab TA session

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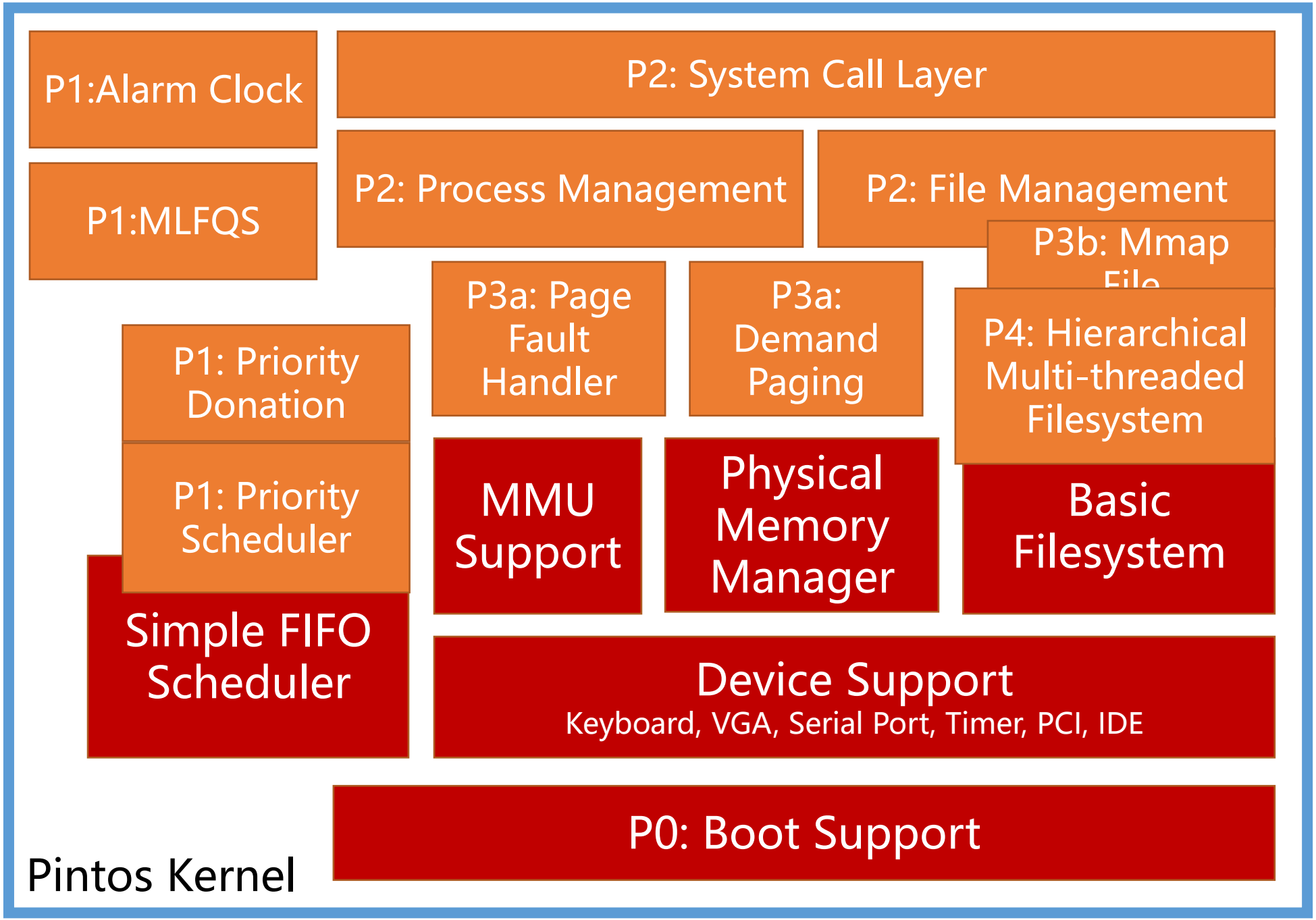
Q : So … what will you do?



P0: Getting Real

P1: Threading

P2: User Programs

P3: Virtual Memory

P4: File System

Students Create

Support Code

Typical workflow:

Lab released
on the Course Website

Read through the lab document

TA session

We are here for Lab 0!

Read through the lab document
and design doc carefully

Design your data structures
and interfaces

Write and Debug code
Pass all the test cases
Submit code zip before code DDL

Answer the questions in design doc
submit it before design doc DDL

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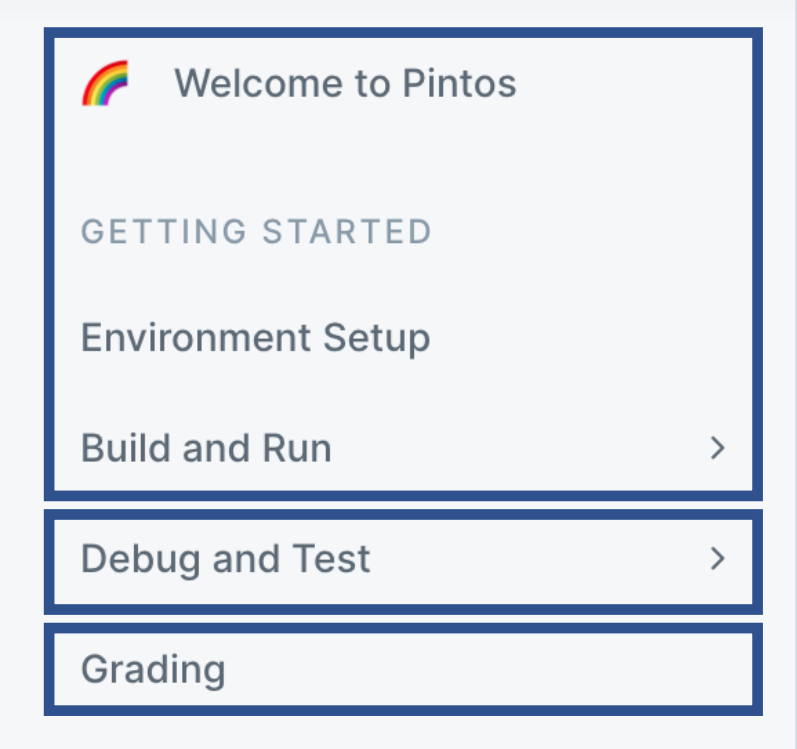
Q :How to survive?

PintosBook long, but helpful



Set up your local development environment.

Look through it and look back if needed.

Important, read it carefully.

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Q :How to survive? PintosBook

PROJECT DESCRIPTION

- Lab0: Getting Real
- Lab1: Threads
- Lab2: User Programs
- Lab3a: Demand Paging
- Lab3b: Mmap Files
- (Optional) Lab4: File Systems

Look through it before each TA Session.

Read it carefully during implementation.

Optional but rewarding Lab4.

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Q :How to survive? PintosBook

APPENDIX

- Code Guide
- 4.4BSD Scheduler
- C Standards
- Project Documentation
- Development Tools
- Bibliography
- Code Browser

PKUOS - Pintos

Pintos source browser for PKU Operating System course

Main Page | Data Structures | Files

File List

Here is a list of all files with brief descriptions:

- src
  - devices
  - examples
  - filesys
  - lib
  - tests
  - threads
  - userprog
  - utils

cts going.

ters.

```
e [IntrList].
```

t List, 2000.

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Q :How to survive?

Your kind TA

Learn to ask questions.

Do not be shy, ask in class, in office hour or in the Piazza.

Office hour: Friday 11:00 a.m -11:59 a.m, Yan Yuan Building 818
Piazza: check https://pku-os.github.io//sp26/

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But … … your TAs are not your ~~personal assistants (or Mr.Deepseek)~~.

- “My program crashed.”
- “What does this error mean?”
- “I failed xxx testcase.”
- “My computer can not boot.”

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Think twice, Ask once.

- How to ask questions the smart way.
- RTFM （Read The Friendly Manual）
- STFW （Search The Friendly Web）
- AI is helpful for reading docs.

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Think twice, Ask once.

- "I encounter xxx under xxx condition."
- "Google says xxx, StackOverflow says xxx, Document says xxx, but yyy."
- "Hey, TA, I found xxx and I think you do not know about it !"

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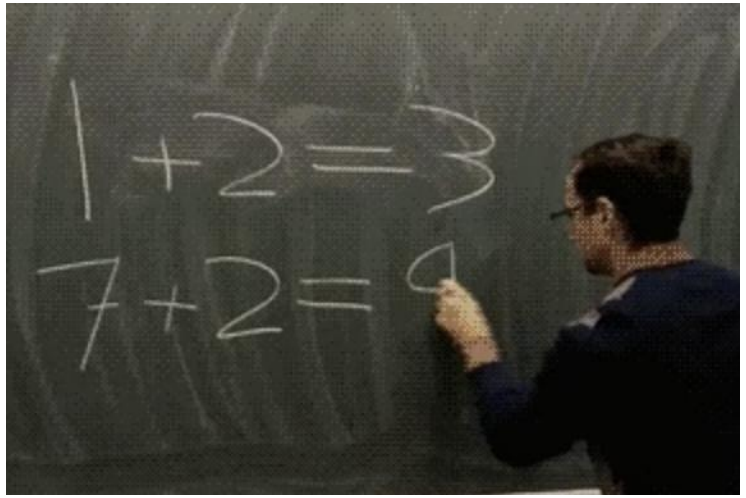
Q :How to survive?

Good habits

awkward, but helpful

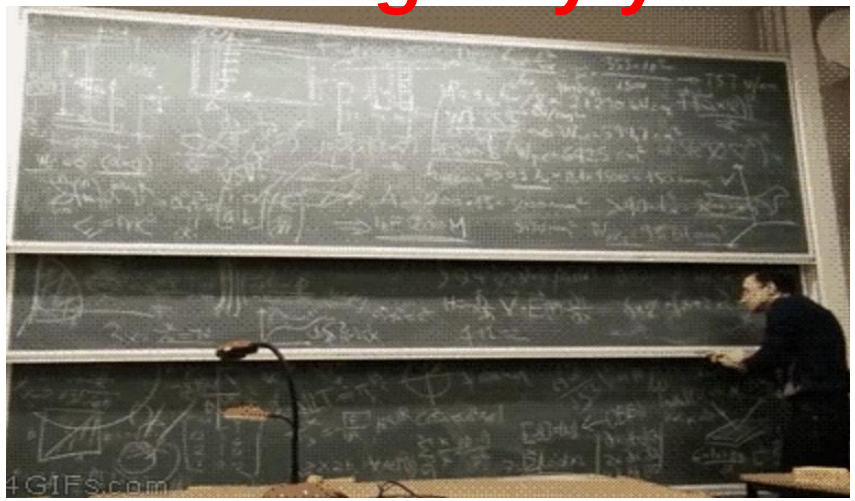
Use Version Control tool —— Git

Learn git by yourself!



Newly written code

A week later



The same code

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How to write good commit message.

Q :How to survive? Good habits

Write concise but good comments.

- Summarize the function in one sentence first.
- Pre-condition: input constraints (You may ASSERT these constraints)
- Post-condition: return value, exception (kernel panic)

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missed comments (you can only omit the comment if the code is self-explained)

2 each, up to 10

Q :How to survive? Good habits

Module and Abstraction.

- A function should (only) do one thing clean
- A function more than 100 LOC warning
- A function more than 200 LOC Something may go wrong

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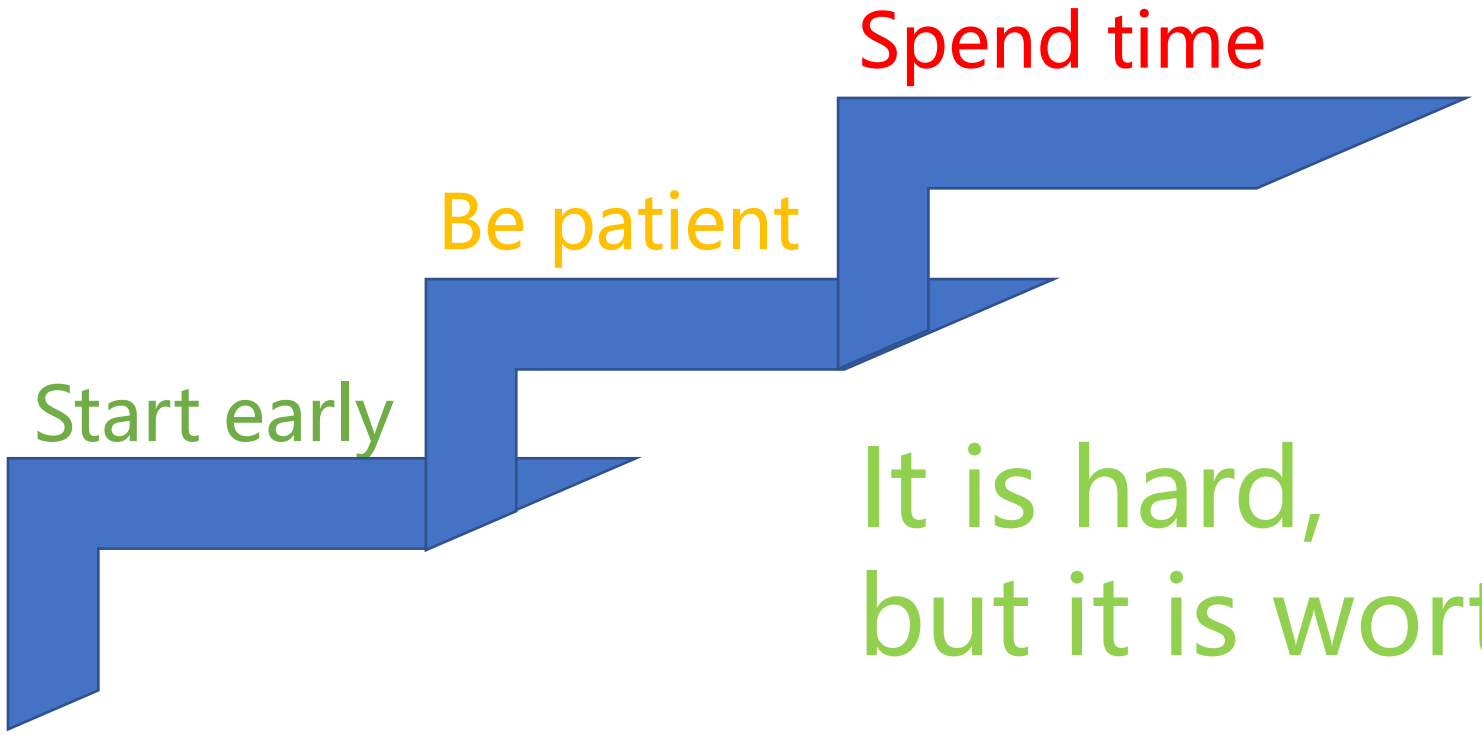
Q :How to survive?

Software Engineering Good Practice

Principles:

- Is your design & code self-explained for your classmates?
  - If not, improve it.
  - Do **NOT** add massive comments to rescue.
- Correctness > simplicity >> performance
- Clean mechanism & interface, fewer preconditions.
- Some bad examples:
  - 调用这个函数之前必须调用另一个函数
  - 存在无法从函数签名中推断的副作用

Q :How to survive?



It is hard,
but it is worth it.

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Q : How about Claude Code

- You can use it.
- But at least design your OS by yourself!

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Lab0 FAQs

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In this Lab, you will be…

- Walking through the **booting** of Pintos
- Try your hands on **debugging** Pintos
- Write your **first line of code** in Pintos: a tiny shell

OS Booting sounds overwhelming?

- All essential information are provided in the PintosBook
- You don't need to master all details; you practice how you learn from new information!

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Tips

- Use old VSCode:
- https://code.visualstudio.com/updates/v1_98

Cutting through the **confusing** jargons…

Docker ENV

You will (assuming you use Docker for running Pintos):

- develop Pintos in a ubuntu:18.04 container (regardless of your host)
- cross-compile Pintos into i386 (Intel 80386) binaries (IA32, 4GB)
- execute Pintos with either QEMU or Bochs, kernel emumators
  - …which come with the BIOS (Basic Input/Output System) firmware that loads a custom OS **bootloader**
    - …which, as part of Pintos, locates and loads the actual **kernel**

Pintos

loader.S

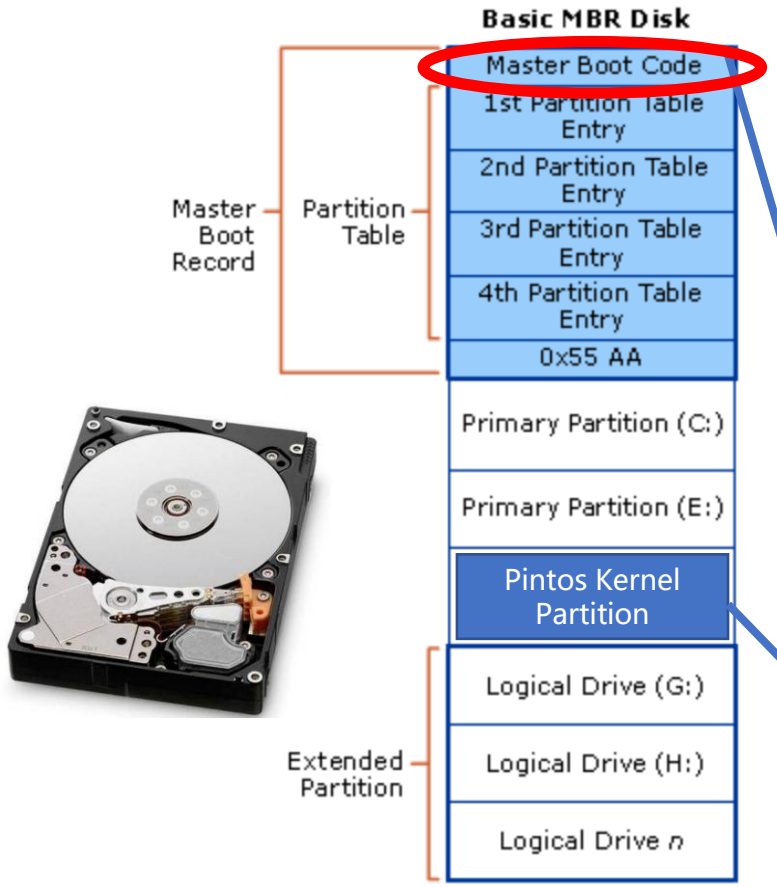
- …which is nothing but an i386 ELF executable, with .text, .data, and an entry point

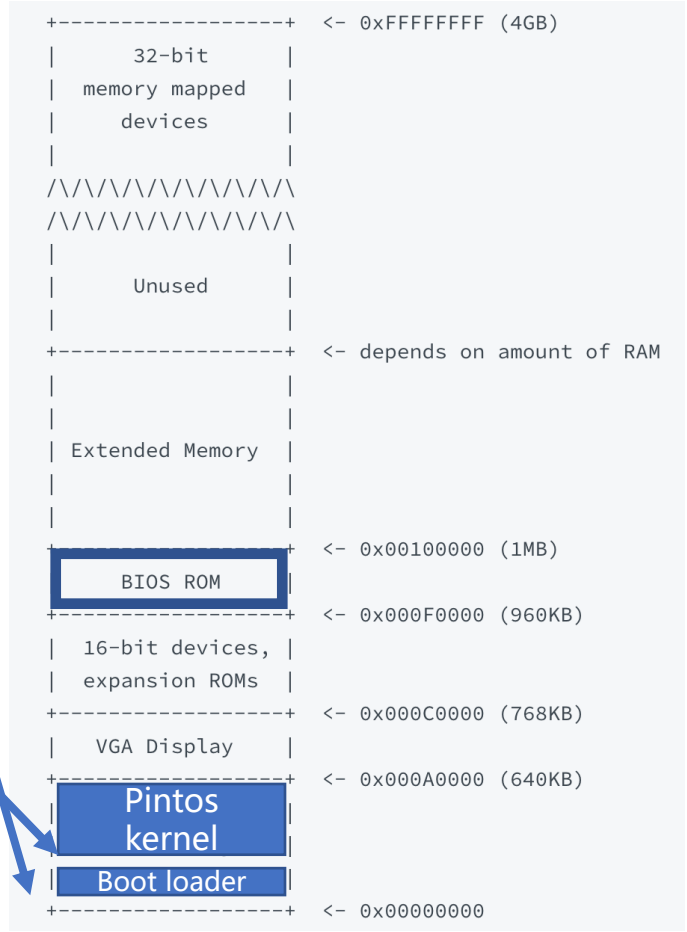
start.S

- …which switches from real mode to protected mode, and calls pintos_init()

Booting Pintos

This MBR code is usually referred to as a boot loader.





Hard-wired by the hardware

The real-world booting process can be much more **complicated**

GRUB (GRand Unified Bootloader),
UEFI (Unified Extensible Firmware Interface), ...

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X86 Mode (history legacy)

X86 Real Mode → Enabled in start.S → X86 Protected Mode

- 16-bit Instructions and Registers

  AX, BX, CX, DX, SI, DI, BP, SP

- 20-bit Memory Address Space (Up to 1MB)

  16-bit segment registers

  CS, DS, SS, ES, FS, GS

  PAddr = SEG << 4 + Operand

- 32-bit Instructions and Registers

  EAX, EBX, ECX, EDX, ESI, EDI, EBP, ESP

- 32-bit Memory Address Space (Up to 4GB)

  Reserved segment registers, but for protection

  Address translation enabled

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And if you really want to know every detail…

- Dig through Makefiles for how the Bootloader (loader.S) and the kernel itself is linked and run
- src/utils/pintos is how you will be running Pintos, which is actually a Perl script that you can try to read and understand

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Conclusion

- Why Pintos?
  - Design and Implementation
  - Read, Design, Write, Debug the code
- What will you do in the projects?
  - Projects Map
  - Typical workflow
- How to survive the projects?
  - PintosBook
  - Ask questions
  - Good habits
  - Good attitude
- Lab0 FAQs: Booting Pintos, X86 mode

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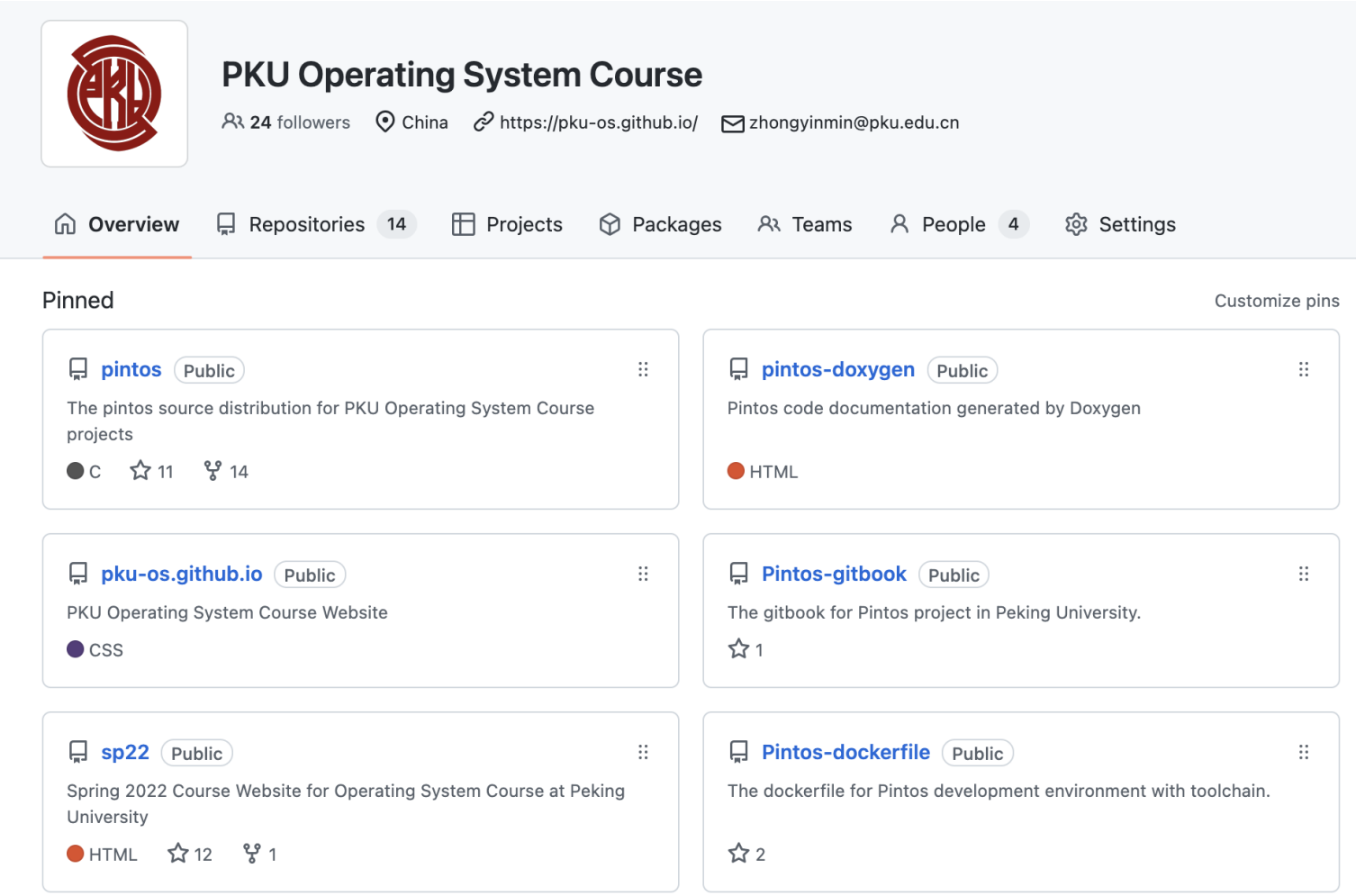
https://github.com/PKU-OS



Learn it,

Master it,

Love it,

and **Join** us.